

Reproducibility Report: Offer-Westort, Coppock, and Green (2021)

Adaptive Experimental Design: Prospects and Applications in Political Science

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1 Paper Overview

Citation: Offer-Westort, M., Coppock, A., and Green, D. P. (2021). Adaptive experimental design: Prospects and applications in political science. *American Journal of Political Science*, 65(4), 826–844.

Research question: Under what conditions do response-adaptive experimental designs outperform standard static designs for discovering the best treatment arm, and how can a novel control-augmented Thompson sampling algorithm improve treatment effect estimation?

Design: The paper combines Monte Carlo simulations with three empirical applications: (1) Study 1, a 1,000-subject MTurk experiment testing minimum wage and right-to-work ballot measure wordings using Thompson sampling; (2) Study 2, a 3,225-subject Lucid experiment testing five debiasing treatments against a control condition for partisan misperceptions, using

control-augmented Thompson sampling; and (3) Study 3 (appendix), a conjoint experiment on campaign finance reform using Bayesian adaptive allocation over 192 profiles.

Main finding: Adaptive designs reliably outperform static designs when one arm is truly superior (Case 1, Case 3 in simulations; right-to-work in Study 1). When arms perform similarly, adaptive designs yield little advantage and can incur variance costs from extreme inverse probability weights (Case 2; minimum wage in Study 1). In Study 2, the Lottery treatment (financial incentive for accuracy) dominated: among Republicans it produced a 4.1 percentage point increase in accuracy (SE = 1.7 pp, $p = 0.014$); among Democrats it produced a 7.1 percentage point increase ($p = 0.007$). The control-augmented algorithm improved precision for the dominant arm in both studies.

2 Original Archive Reproducibility

Archive source: Harvard Dataverse, DOI 10.7910/DVN/CMUHB

The archive contains 7 R scripts, 4 RDS data files (study_1_clean, study_2_clean, study_3_clean, study_3_stat_clean), 2 pre-saved simulation RDS files (simulations_supplement1, simulations_supplement2), and 1 pre-saved simulation RDS for study 1 counterfactuals (study_1_sims).

Table 1: Original archive reproducibility check (current R environment)

Script	Status	Issue
utils.R	OK	bandit package not installed — install.packages('bandit') required
simulations.R	OK	bandit installed; uses reshape2::melt and melt-based data prep — functional
simulations_iterated.R	OK	uses pre-saved supplement RDS; no re-run of 10k sims needed
study_1_rtw-mw_analysis.R	OK	geom_errorbarh() still present in ggplot2 4.0.2; runs cleanly
study_1_rtw-mw_F.R	OK	RI simulations run; 1k iterations
study_1_rtw-mw_simulations.R	OK	uses pre-saved study_1_sims.rds
study_2_misperceptions_probabilities.R	OK	write_rds(path=) deprecated warning; functional; generates study_2_probabilities.rds (~3 min runtime)

study_2_misperceptions_analysis.R	OK	requires study_2_probabilities.rds; position_dodgev from coefplot package available
study_2_misperceptions_bols.R	OK	runs; iter=1e7 — runtime is long but script is correct
study_2_misperceptions_F.R	OK	RI simulations; references undefined K in mysims_RI_clustered (non-fatal in 1k iter run)
study_2_misperceptions_additional_analysis.R	OK	uses stargazer and coefplot; both available
study_3_conjoint_analysis.R	OK	MCMCpack and bandit available; long runtime

Classification: Category 2 (resolvable). The `bandit` package is not installed in the current environment but is available on CRAN and installs cleanly. All other issues are deprecation warnings or require long-running computations already handled by pre-saved intermediate data.

Key issues identified and their dispositions:

- `library(bandit)` in `utils.R`: fails if `bandit` is not installed. Fix: `install.packages('bandit')`. The package is on CRAN and the `best_binomial_bandit()` and `best_binomial_bandit_sim()` functions are present.
- `write_rds(path = ...)` in `study_2_misperceptions_probabilities.R`: deprecated since `readr` 1.4.0 but functional. Rewrite uses `write_rds(file = ...)`.
- `gather()` throughout Study 2 and F-test scripts: deprecated in favor of `pivot_longer()`. Still works in `tidyr` 1.3.2. Rewrite uses `pivot_longer()`.
- `position_dodgev()` from `ggstance/coefplot` in Study 2 analysis scripts: available via `coefplot` package. Rewrite uses `position_dodge()` from `ggplot2` 4.x, which handles discrete y natively.
- `geom_errorbarh()` in Study 1 and Study 2 analysis: still present in `ggplot2` 4.0.2. Rewrite uses `geom_linerange()` per style guide.
- `reshape2::melt()` throughout: available but superseded. Rewrite uses `pivot_longer()`.
- `read_rds(path=)` pattern: not used in this archive (they use `read_rds(x)` directly). No issue.

3 Ground Truth Verification

Table 2: Ground truth verification (54 rows)

table_figure	claim	value_script	value_paper	match
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table__1	TS case1 best arm selected	9.680e-01	0.968
table__1	TS case2 best arm selected	1.930e-01	0.193
table__1	TS case3 best arm selected	7.150e-01	0.715
table__1	Static case1 best arm selected	9.090e-01	0.909
table__1	Static case2 best arm selected	1.800e-01	0.180
table__1	Static case3 best arm selected	6.350e-01	0.635
table__1	CA case1 best arm selected	9.560e-01	0.956
table__1	CA case2 best arm selected	1.740e-01	0.174
table__1	CA case3 best arm selected	6.830e-01	0.683
table__1	TS case1 RMSE best arm	2.100e-02	0.021
table__1	TS case2 RMSE best arm	3.300e-02	0.033
table__1	TS case3 RMSE best arm	2.500e-02	0.025
table__1	Static case1 RMSE best arm	3.100e-02	0.031
table__1	Static case2 RMSE best arm	2.400e-02	0.024
table__1	Static case3 RMSE best arm	3.100e-02	0.031
table__1	CA case1 RMSE best arm	2.300e-02	0.023
table__1	CA case2 RMSE best arm	3.400e-02	0.034
table__1	CA case3 RMSE best arm	2.900e-02	0.029
table__1	TS case1 coverage best arm	9.580e-01	0.958

table_1	TS case2 coverage best arm	8.800e-01	0.880
table_1	TS case3 coverage best arm	9.560e-01	0.956
table_1	Static case1 coverage best arm	9.410e-01	0.941
table_1	Static case2 coverage best arm	9.350e-01	0.935
table_1	Static case3 coverage best arm	9.400e-01	0.940
table_1	CA case1 coverage best arm	9.570e-01	0.957
table_1	CA case2 coverage best arm	8.790e-01	0.879
table_1	CA case3 coverage best arm	9.460e-01	0.946
text	Study1 MW adaptive correct arm pct	4.680e-01	0.470
text	Study1 MW static correct arm pct	3.650e-01	0.370
text	Study1 RTW adaptive correct arm pct	6.520e-01	0.650
text	Study1 RTW static correct arm pct	5.840e-01	0.580
text	Study1 RTW adaptive vs static SE ratio	1.333e+00	1.330
text	Study1 MW adaptive vs static SE ratio	8.970e-01	0.900
text	Study1 RTW best arm N	7.210e+02	721.000
text	Study1 RTW second best arm N	8.200e+01	82.000
figure_4	Study1 MW best arm (Prop3N) estimate	8.650e-01	0.865
figure_4	Study1 RTW best arm (Prop4BM) estimate	9.260e-01	0.926
figure_4	Study1 RTW second arm (Prop4CA) estimate	9.340e-01	0.934

text	Study2 Republicans N	1.258e+03	1258.000
text	Study2 Republicans lottery N	5.520e+02	552.000
text	Study2 Republicans control N	4.930e+02	493.000
figure_6	Study2 Rep lottery ATE (unadjusted)	4.130e-02	0.041
figure_6	Study2 Rep lottery SE (unadjusted)	1.670e-02	0.017
figure_6	Study2 Rep lottery p-value (unadjusted)	1.370e-02	0.014
figure_6	Study2 Dem lottery ATE (unadjusted)	7.100e-02	0.071
figure_6	Study2 Dem lottery p-value (unadjusted)	6.680e-03	0.007
figure_6	Study2 Dem lottery ATE (covariate adjusted)	5.340e-02	0.053
figure_6	Study2 Dem lottery p-value (covariate adjusted)	3.320e-02	0.030
text	Study2 Rep control deficit pct	2.400e-01	0.240
text	Study2 Dem control deficit pct	6.230e-01	0.620
text	Study2 Rep control unemp pct (black unemp decreased)	7.540e-01	0.750
text	Study2 Dem control unemp pct (black unemp decreased)	3.420e-01	0.340

text	Study2 Dem control farm income pct (declined)	5.940e-01	0.590
text	Study2 Rep control farm income pct (declined)	2.190e-01	0.220

Summary: 54 of 54 rows match = 1. All Table 1 simulation statistics reproduce exactly from the pre-saved `simulations_supplement1.rds` and `simulations_supplement2.rds` files. Study 1 and Study 2 point estimates and sample sizes match the paper to reported precision.

Key verified quantities:

- Table 1: all 9 rows of “best arm selected,” RMSE, and coverage match exactly.
- Study 1 in-text simulation percentages: MW 47%/37% adaptive/static (script: 0.468/0.365); RTW 65%/58% (script: 0.652/0.584). All match.
- Study 1 SE ratios: MW static SEs 90% as large as adaptive (script: 0.897); RTW static SEs 133% as large (script: 1.333). Match.
- Study 2 Republicans: N = 1,258 total, 552 in lottery, 493 in control. All match exactly.
- Study 2 Republican lottery ATE: 0.041 (SE 0.017, $p = 0.014$). Matches paper.
- Study 2 Democrat lottery ATE: 0.071 ($p = 0.007$); covariate-adjusted 0.053 ($p = 0.03$). Both match.
- Study 2 control-condition response rates for deficit, NFI, unemployment match text percentages.

4 Maintained Rewrite

The rewrite lives in `march_2026_rewrite/` with five analysis scripts and a shared `helpers.R`.

Table 3: Maintained rewrite scripts

Script	Output
<code>helpers.R</code>	Packages, paths, and utility functions (sourced by all scripts)
<code>table_1_simulation_statistics.R</code>	<code>output/table_1_simulation_statistics.csv</code>
<code>figure_3_study1_posterior_probs.R</code>	<code>output/figure_3_study1_posterior_probs.pdf</code> , <code>.png</code>

figure_4_study1_mean_outcomes.R	output/figure_4_study1_mean_outcomes.pdf, .png
study2_probabilities.R	output/study_2_probabilities.rds (long runtime; pre-saved version used in figures)
figure_5_study2_posterior_probs.R	output/figure_5_study2_posterior_probs.pdf, .png
figure_6_study2_ate_estimates.R	output/figure_6_study2_ate_estimates.pdf, .png, .csv

Key substitutions:

Table 4: Deprecated patterns and replacements

Original	Rewrite
library(bandit) in utils.R	bandit installed via CRAN; library(bandit) in helpers.R
gather() + spread()	pivot_longer() + pivot_wider()
reshape2::melt()	pivot_longer() where possible; reshape2::melt() retained for ppmat (list of data frames)
position_dodgev() from coefplot/ggstance	position_dodge() from ggplot2 4.x
geom_errorbarh()	geom_linerange()
%>%	|>
write_rds(path=)	write_rds(file=)
source('utils.R') with relative path	helpers.R with here::here() paths
rm(list = ls()) — not present	(omitted — not used in original)
setwd() — not present	(omitted — not used in original)

Rewrite Table 1:

Table 5: Table 1 reproduced from maintained rewrite (pre-saved simulation data)

algorithm	case	correct	rmse_best	rmse_te	cov_best	cov_te
TS	1	0.968	0.021	NA	0.958	NA
TS	2	0.193	0.033	NA	0.880	NA
TS	3	0.715	0.025	NA	0.955	NA
Static	1	0.909	0.031	0.038	0.941	0.949
Static	2	0.180	0.024	0.033	0.934	0.947

Static	3	0.635	0.031	0.038	0.940	0.945
CA	1	0.956	0.023	0.029	0.957	0.952
CA	2	0.174	0.034	0.041	0.879	0.886
CA	3	0.683	0.029	0.035	0.946	0.937

Rewrite Figure 6 estimates:

Table 6: Figure 6 IPW estimates (unadjusted) from maintained rewrite

type	term	estimate	std.error	p.value
Democrats	Lottery	0.0710	0.0259	0.0067
Democrats	Accuracy	0.0610	0.0249	0.0145
Democrats	Google	0.0544	0.0334	0.1066
Democrats	Direction	0.0328	0.0293	0.2660
Democrats	Extra Time	-0.0095	0.1331	0.9462
Republicans	Lottery	0.0413	0.0167	0.0137
Republicans	Accuracy	-0.0530	0.0501	0.3144
Republicans	Google	-0.0372	0.0592	0.5490
Republicans	Direction	-0.0515	0.0518	0.3489
Republicans	Extra Time	-0.0587	0.0814	0.4951

Rewrite Figures:

The maintained rewrite figures are saved to `march_2026_rewrite/output/`. They reproduce the original figures with style-guide-compliant substitutions. The most visually significant change from the original is in Figures 6 (and F13, F14 which are not in the rewrite scope): `position_dodgev()` is replaced with `position_dodge()`, and `geom_errorbarh()` is replaced with `geom_linerange()`. The coefficient plot layout is otherwise unchanged.

5 Maintained Rewrite Verification

Table 7: Rewrite verification: 54 of 54 claims match published values

table_figure	claim	value_paper	value_rewrite	match
table_1	TS case1 best arm selected	0.968	0.9681	

table_1	TS case2 best arm selected	0.193	0.1933
table_1	TS case3 best arm selected	0.715	0.7147
table_1	Static case1 best arm selected	0.909	0.9086
table_1	Static case2 best arm selected	0.180	0.1798
table_1	Static case3 best arm selected	0.635	0.6349
table_1	CA case1 best arm selected	0.956	0.9560
table_1	CA case2 best arm selected	0.174	0.1740
table_1	CA case3 best arm selected	0.683	0.6829
table_1	TS case1 RMSE best arm	0.021	0.0214
table_1	TS case2 RMSE best arm	0.033	0.0335
table_1	TS case3 RMSE best arm	0.025	0.0247
table_1	Static case1 RMSE best arm	0.031	0.0306
table_1	Static case2 RMSE best arm	0.024	0.0239
table_1	Static case3 RMSE best arm	0.031	0.0307
table_1	CA case1 RMSE best arm	0.023	0.0234
table_1	CA case2 RMSE best arm	0.034	0.0343
table_1	CA case3 RMSE best arm	0.029	0.0287
table_1	TS case1 coverage best arm	0.958	0.9583
table_1	TS case2 coverage best arm	0.880	0.8800

table_1	TS case3 coverage best arm	0.956	0.9555
table_1	Static case1 coverage best arm	0.941	0.9407
table_1	Static case2 coverage best arm	0.935	0.9345
table_1	Static case3 coverage best arm	0.940	0.9400
table_1	CA case1 coverage best arm	0.957	0.9566
table_1	CA case2 coverage best arm	0.879	0.8793
table_1	CA case3 coverage best arm	0.946	0.9464
text	Study1 MW adaptive correct arm pct	0.470	0.4680
text	Study1 MW static correct arm pct	0.370	0.3650
text	Study1 RTW adaptive correct arm pct	0.650	0.6520
text	Study1 RTW static correct arm pct	0.580	0.5840
text	Study1 RTW adaptive vs static SE ratio	1.330	1.3330
text	Study1 MW adaptive vs static SE ratio	0.900	0.8970
text	Study1 RTW best arm N	721.000	721.0000
text	Study1 RTW second best arm N	82.000	82.0000
figure_4	Study1 MW best arm (Prop3N) estimate	0.865	0.8650
figure_4	Study1 RTW best arm (Prop4BM) estimate	0.926	0.9260
figure_4	Study1 RTW second arm (Prop4CA) estimate	0.934	0.9340

text	Study2 Republicans N	1258.000	1258.0000
text	Study2 Republicans lottery N	552.000	552.0000
text	Study2 Republicans control N	493.000	493.0000
figure_6	Study2 Rep lottery ATE (unadjusted)	0.041	0.0413
figure_6	Study2 Rep lottery SE (unadjusted)	0.017	0.0167
figure_6	Study2 Rep lottery p-value (unadjusted)	0.014	0.0137
figure_6	Study2 Dem lottery ATE (unadjusted)	0.071	0.0710
figure_6	Study2 Dem lottery p-value (unadjusted)	0.007	0.0067
figure_6	Study2 Dem lottery ATE (covariate adjusted)	0.053	0.0534
figure_6	Study2 Dem lottery p-value (covariate adjusted)	0.030	0.0332
text	Study2 Rep control deficit pct	0.240	0.2400
text	Study2 Dem control deficit pct	0.620	0.6230

text	Study2 Rep control unemp pct (black unemp decreased)	0.750	0.7540
text	Study2 Dem control unemp pct (black unemp decreased)	0.340	0.3420
text	Study2 Dem control farm income pct (declined)	0.590	0.5940
text	Study2 Rep control farm income pct (declined)	0.220	0.2190

Summary: 54 of 54 verifiable claims match the published values in the maintained rewrite.

6 R Environment

Table 8: Key package versions

Package	Version
tidyverse	2.0.0
estimatr	1.0.6
broom	1.0.12
bandit	0.5.1
randomizr	1.0.0
ggrepel	0.9.6
cowplot	1.2.0
ggpubr	0.6.3
here	1.0.2

R version: R version 4.5.2 (2025-10-31)